Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Algebraic-Logarithmic Definite Integral

Evaluate the definite integral:

$$\int_0^1 \frac{(x+\ln(x)-1)^2}{(\ln(x))(x-1)}\,dx$$

Solution: Let the integral be denoted by I. We can expand the integrand by treating the numerator as the square of the sum $(x-1)+\ln(x)$.

$$\frac{((x-1)+\ln(x))^2}{(x-1)\ln(x)} = \frac{(x-1)^2+2(x-1)\ln(x)+(\ln(x))^2}{(x-1)\ln(x)}$$

Distributing the denominator across the three terms in the numerator, we obtain:

$$\frac{(x-1)^2}{(x-1)\ln(x)} + \frac{2(x-1)\ln(x)}{(x-1)\ln(x)} + \frac{(\ln(x))^2}{(x-1)\ln(x)} = \frac{x-1}{\ln(x)} + 2 + \frac{\ln(x)}{x-1}$$

Now, we can split the original integral I into three separate integrals:

$$I = \int_0^1 \frac{x-1}{\ln(x)}\,dx + \int_0^1 2\,dx + \int_0^1 \frac{\ln(x)}{x-1}\,dx$$

We evaluate each integral individually:

- The first integral is a well-known parameterized integral. Recall that $\int_0^1 \frac{x^a-1}{\ln(x)}\,dx = \ln(a+1)$ for $a>-1$. Setting $a=1$, we get:

$$\int_0^1 \frac{x-1}{\ln(x)}\,dx = \ln(1+1) = \ln(2)$$

- The second integral is trivial:

$$\int_0^1 2\,dx = 2(1-0) = 2$$

- For the third integral, we apply the substitution $u=1-x$, which gives $du=-dx$. The limits of integration swap, which is cancelled by the negative sign:

$$\int_0^1 \frac{\ln(x)}{x-1}\,dx = \int_0^1 \frac{\ln(1-u)}{-u}\,du = -\int_0^1 \frac{\ln(1-u)}{u}\,du$$

  This is the integral representation of the Dilogarithm function, specifically $\mathrm{Li}_2(1)$, which evaluates to $\frac{\pi^2}{6}$:

$$\int_0^1 \frac{\ln(x)}{x-1}\,dx = \frac{\pi^2}{6}$$

Summing the evaluated parts yields the exact value of the integral I:

$$I = \ln(2) + 2 + \frac{\pi^2}{6}$$

1.2 Hard Integral

Trigonometric-Logarithmic Integral

Evaluate the definite integral:

$$\int_0^{\pi} \frac{\cos\left(\frac{x}{2}\right)}{\sqrt{\sin(x)}} \ln\left(\tan\left(\frac{x}{2}\right)\right) dx$$

Solution: Let the target integral be denoted by I. We use the tangent half-angle substitution (Weierstrass substitution). Let $u = \tan\left(\frac{x}{2}\right)$. Then, the differential is $dx = \frac{2}{1+u^2}\, du$. The relevant trigonometric functions transform as follows:

$$\sin(x) = \frac{2u}{1+u^2}, \quad \cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+u^2}}$$

The limits of integration change from $x = 0 \implies u = 0$ to $x = \pi \implies u \to \infty$. Substituting these into the integral I:

$$I = \int_0^{\infty} \frac{\frac{1}{\sqrt{1+u^2}}}{\sqrt{\frac{2u}{1+u^2}}} \ln(u) \frac{2}{1+u^2}\, du$$

Simplifying the integrand:

$$I = \int_0^{\infty} \frac{1}{\sqrt{1+u^2}} \frac{\sqrt{1+u^2}}{\sqrt{2u}} \ln(u) \frac{2}{1+u^2}\, du = \sqrt{2}\int_0^{\infty} \frac{\ln(u)}{\sqrt{u}(1+u^2)}\, du$$

Next, we apply the substitution $u = t^{1/2}$, which means $du = \frac{1}{2}t^{-1/2}\, dt$. The limits remain 0 to ∞.

$$I = \sqrt{2}\int_0^{\infty} \frac{\ln(t^{1/2})}{t^{1/4}(1+t)} \frac{1}{2} t^{-1/2}\, dt = \frac{\sqrt{2}}{4}\int_0^{\infty} \frac{\ln(t)}{t^{3/4}(1+t)}\, dt$$

We recognize this integral as the derivative of a known parametric Beta function integral. Recall that for $0 < a < 1$:

$$J(a) = \int_0^{\infty} \frac{t^{a-1}}{1+t}\, dt = \frac{\pi}{\sin(\pi a)}$$

Differentiating both sides with respect to a introduces the natural logarithm:

$$J'(a) = \int_0^{\infty} \frac{t^{a-1}\ln(t)}{1+t}\, dt = \frac{d}{da}\left(\frac{\pi}{\sin(\pi a)}\right) = -\frac{\pi^2\cos(\pi a)}{\sin^2(\pi a)}$$

In our case, the exponent of t in the denominator is $3/4$, which means $a - 1 = -3/4$, so $a = 1/4$. We substitute $a = 1/4$ into our derivative formula:

$$\int_0^{\infty} \frac{t^{-3/4}\ln(t)}{1+t}\, dt = -\frac{\pi^2\cos(\pi/4)}{\sin^2(\pi/4)}$$

Given $\cos(\pi/4) = \frac{\sqrt{2}}{2}$ and $\sin(\pi/4) = \frac{\sqrt{2}}{2}$:

$$-\frac{\pi^2\left(\frac{\sqrt{2}}{2}\right)}{\left(\frac{\sqrt{2}}{2}\right)^2} = -\frac{\pi^2\frac{\sqrt{2}}{2}}{\frac{1}{2}} = -\sqrt{2}\pi^2$$

Finally, substitute this value back into the expression for I:

$$I = \frac{\sqrt{2}}{4}\left(-\sqrt{2}\pi^2\right) = -\frac{2\pi^2}{4} = -\frac{\pi^2}{2}$$

The exact value is:

$$-\frac{\pi^2}{2}$$

Differentials

2.1 Medium Differential

Second-Order Initial Value Problem

Solve the second-order differential equation:

$$y'' + 4y = 0$$

Given the initial conditions $y(0) = 0$ and $y'(0) = 1$, find the exact value of $y\left(\frac{\pi}{3}\right)$.

Solution: We have a second-order linear homogeneous differential equation with constant coefficients. We start by formulating its characteristic equation:

$$r^2 + 4 = 0 \implies r^2 = -4 \implies r = \pm 2i$$

Since the roots are purely imaginary, the general solution takes the form of a linear combination of sine and cosine functions:

$$y(x) = C_1 \cos(2x) + C_2 \sin(2x)$$

We apply the first initial condition $y(0) = 0$:

$$y(0) = C_1 \cos(0) + C_2 \sin(0) = C_1(1) + C_2(0) = C_1 = 0$$

With $C_1 = 0$, our solution simplifies to $y(x) = C_2 \sin(2x)$. We now differentiate this to apply the second condition:

$$y'(x) = 2C_2 \cos(2x)$$

Applying the condition $y'(0) = 1$:

$$y'(0) = 2C_2 \cos(0) = 2C_2 = 1 \implies C_2 = \frac{1}{2}$$

The particular solution that satisfies the given initial conditions is therefore:

$$y(x) = \frac{1}{2} \sin(2x)$$

To find the required value, we evaluate this function at $x = \frac{\pi}{3}$:

$$y\left(\frac{\pi}{3}\right) = \frac{1}{2} \sin\left(2 \cdot \frac{\pi}{3}\right) = \frac{1}{2} \sin\left(\frac{2\pi}{3}\right)$$

Since $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$:

$$y\left(\frac{\pi}{3}\right) = \frac{1}{2}\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4}$$

The exact value is:

$$\frac{\sqrt{3}}{4}$$

2.2 Hard Differential

First-Order Linear Differential Equation with Series Coefficient

Find the exact value of $y(\pi)$, given the first-order linear differential equation:

$$y' + \frac{y}{\sqrt{x^2-1}} = \left(\sum_{n=1}^{\infty} \frac{[\cos^2(n)]+1}{n^4}\right) \cdot \frac{x+\sqrt{x^2-1}}{\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right)}$$

with the initial condition $y(1) = 0$. Note that $[\cdot]$ denotes the integer part of a value, and the domain is $x \geq 1$.

Solution: Let us first evaluate the complex constant term on the right-hand side. For any integer $n \geq 1$, the value of $\cos^2(n)$ lies strictly in the interval $(0, 1)$ because π is irrational, meaning n can never be an exact multiple of π. Therefore, the integer part $[\cos^2(n)] = 0$. The infinite sum simplifies to:

$$\sum_{n=1}^{\infty} \frac{0+1}{n^4} = \sum_{n=1}^{\infty} \frac{1}{n^4} = \zeta(4) = \frac{\pi^4}{90}$$

For the denominator, we use Euler's Reflection Formula $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$:

$$\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right) = \frac{\pi}{\sin\left(\frac{\pi}{3}\right)} = \frac{\pi}{\frac{\sqrt{3}}{2}} = \frac{2\pi}{\sqrt{3}}$$

Thus, the constant multiplier C is:

$$C = \frac{\frac{\pi^4}{90}}{\frac{2\pi}{\sqrt{3}}} = \frac{\sqrt{3}\pi^3}{180}$$

The differential equation becomes:

$$y' + \frac{1}{\sqrt{x^2-1}}y = C(x+\sqrt{x^2-1})$$

This is a standard first-order linear ordinary differential equation. We find the integrating factor $\mu(x)$:

$$\mu(x) = \exp\left(\int \frac{1}{\sqrt{x^2-1}}dx\right) = \exp(\cosh^{-1}(x)) = x+\sqrt{x^2-1}$$

Multiplying both sides by the integrating factor gives:

$$\frac{d}{dx}\left[y\left(x+\sqrt{x^2-1}\right)\right] = C\left(x+\sqrt{x^2-1}\right)^2$$

We substitute $u = x+\sqrt{x^2-1}$. The derivative with respect to x is $\frac{du}{dx} = 1 + \frac{x}{\sqrt{x^2-1}} = \frac{u}{\sqrt{x^2-1}}$. Alternatively, we can express the integral directly. Using the hyperbolic substitution $x = \cosh(\theta)$, we have $dx = \sinh(\theta)d\theta$ and $x+\sqrt{x^2-1} = e^{\theta}$.

$$\int \left(x+\sqrt{x^2-1}\right)^2 dx = \int e^{2\theta}\sinh(\theta)\,d\theta = \int e^{2\theta}\left(\frac{e^{\theta}-e^{-\theta}}{2}\right) d\theta$$

$$= \frac{1}{2}\int \left(e^{3\theta} - e^{\theta}\right)\, d\theta = \frac{1}{6}e^{3\theta} - \frac{1}{2}e^{\theta} = \frac{1}{6}\left(x + \sqrt{x^2-1}\right)^3 - \frac{1}{2}\left(x + \sqrt{x^2-1}\right)$$

So the general solution is:

$$y\left(x + \sqrt{x^2-1}\right) = C\left[\frac{1}{6}\left(x + \sqrt{x^2-1}\right)^3 - \frac{1}{2}\left(x + \sqrt{x^2-1}\right)\right] + K$$

Using the initial condition $y(1) = 0$. At $x = 1$, $x + \sqrt{x^2-1} = 1$:

$$0 \cdot 1 = C\left[\frac{1}{6}(1)^3 - \frac{1}{2}(1)\right] + K \implies K = -C\left(\frac{1}{6} - \frac{1}{2}\right) = -C\left(-\frac{1}{3}\right) = \frac{C}{3}$$

Substitute K back and divide by $u = x + \sqrt{x^2-1}$:

$$y(x) = C\left[\frac{1}{6}\left(x + \sqrt{x^2-1}\right)^2 - \frac{1}{2} + \frac{1}{3\left(x + \sqrt{x^2-1}\right)}\right]$$

Rationalizing the last term using $\frac{1}{x+\sqrt{x^2-1}} = x - \sqrt{x^2-1}$:

$$y(x) = \frac{\sqrt{3}\pi^3}{180}\left[\frac{1}{6}\left(x + \sqrt{x^2-1}\right)^2 - \frac{1}{2} + \frac{1}{3}\left(x - \sqrt{x^2-1}\right)\right]$$

Evaluating at $x = \pi$:

$$y(\pi) = \frac{\sqrt{3}\pi^3}{180}\left[\frac{1}{6}\left(\pi + \sqrt{\pi^2-1}\right)^2 - \frac{1}{2} + \frac{1}{3}\left(\pi - \sqrt{\pi^2-1}\right)\right]$$

Derivatives

3.1 Medium Derivative

Complex Trigonometric Quotient Derivative

Let $f(x) = \frac{\sin(\ln(x^2))\cos(\ln(x^2))}{\sin(\ln(x))\tan(x) - 2\sin^3(\ln(x))\tan(x)}$. Find $f'(1)$.

Solution: Before differentiating, we can greatly simplify the function $f(x)$ using trigonometric identities. First, we look at the numerator. We know $\ln(x^2) = 2\ln(x)$. Using the double-angle identity for sine, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$:

$$\text{Numerator} = \sin(2\ln(x))\cos(2\ln(x)) = \frac{1}{2}\sin(4\ln(x))$$

Alternatively, we can express it strictly in terms of $\ln(x)$:

$$\text{Numerator} = 2\sin(\ln(x))\cos(\ln(x))\cos(2\ln(x))$$

Next, we factor the denominator:

$$\text{Denominator} = \tan(x)\sin(\ln(x))\left[1 - 2\sin^2(\ln(x))\right]$$

We recognize the expression in the brackets as the double-angle identity for cosine: $1 - 2\sin^2(\theta) = \cos(2\theta)$. Thus:

$$\text{Denominator} = \tan(x)\sin(\ln(x))\cos(2\ln(x))$$

Now, we substitute the simplified numerator and denominator back into $f(x)$:

$$f(x) = \frac{2\sin(\ln(x))\cos(\ln(x))\cos(2\ln(x))}{\tan(x)\sin(\ln(x))\cos(2\ln(x))}$$

Assuming x is in a domain where the denominator is non-zero, we cancel the common terms $\sin(\ln(x))$ and $\cos(2\ln(x))$:

$$f(x) = \frac{2\cos(\ln(x))}{\tan(x)} = 2\cos(\ln(x))\cot(x)$$

Now, we differentiate this simplified function using the product rule and chain rule:

$$f'(x) = 2\left[\frac{d}{dx}(\cos(\ln(x))) \cdot \cot(x) + \cos(\ln(x)) \cdot \frac{d}{dx}(\cot(x))\right]$$

$$f'(x) = 2\left[-\sin(\ln(x)) \cdot \frac{1}{x} \cdot \cot(x) + \cos(\ln(x)) \cdot (-\csc^2(x))\right]$$

We evaluate this derivative at $x = 1$. Note that $\ln(1) = 0$, $\sin(0) = 0$, and $\cos(0) = 1$:

$$f'(1) = 2\left[-\sin(0) \cdot \frac{1}{1} \cdot \cot(1) - \cos(0) \cdot \csc^2(1)\right]$$

$$f'(1) = 2\left[0 - 1 \cdot \csc^2(1)\right] = -2\csc^2(1)$$

The exact value is:

$$-2\csc^2(1)$$

3.2 Hard Derivative

Higher-Order Derivative at Origin

Let $f(x) = x^4 \tan(x^3) - x \ln(1 + x^2)$. Find $f'''''(0)$.

Solution: We need to find the fifth derivative of $f(x)$ evaluated at $x = 0$. Using the Maclaurin series expansion definition, we know that for a smooth function:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Therefore, $f^{(5)}(0)$ is exactly 5! times the coefficient of the x^5 term in the Maclaurin series of $f(x)$. Let us find the Maclaurin series expansions for the two components of $f(x)$.

First, consider the term $x^4 \tan(x^3)$. We know the standard Maclaurin series for tangent is:

$$\tan(u) = u + \frac{1}{3}u^3 + \frac{2}{15}u^5 + \mathcal{O}(u^7)$$

Substituting $u = x^3$:

$$\tan(x^3) = x^3 + \frac{1}{3}x^9 + \mathcal{O}(x^{15})$$

Multiplying by x^4:

$$x^4 \tan(x^3) = x^4 \left(x^3 + \frac{1}{3}x^9 + \dots \right) = x^7 + \frac{1}{3}x^{13} + \dots$$

This term consists strictly of powers of x degree 7 and higher. The coefficient of x^5 in this part is exactly 0.

Second, consider the term $x \ln(1 + x^2)$. We know the standard Maclaurin series for the natural logarithm is:

$$\ln(1 + u) = u - \frac{1}{2}u^2 + \frac{1}{3}u^3 - \frac{1}{4}u^4 + \mathcal{O}(u^5)$$

Substituting $u = x^2$:

$$\ln(1 + x^2) = x^2 - \frac{1}{2}x^4 + \frac{1}{3}x^6 - \mathcal{O}(x^8)$$

Multiplying by x:

$$x \ln(1 + x^2) = x \left(x^2 - \frac{1}{2}x^4 + \frac{1}{3}x^6 - \dots \right) = x^3 - \frac{1}{2}x^5 + \frac{1}{3}x^7 - \dots$$

Combining the two series to form the full expansion of $f(x)$:

$$f(x) = (x^7 + \dots) - \left(x^3 - \frac{1}{2}x^5 + \frac{1}{3}x^7 - \dots \right) = -x^3 + \frac{1}{2}x^5 + \frac{2}{3}x^7 + \dots$$

The coefficient of the x^5 term in the series of $f(x)$ is exactly $\frac{1}{2}$. Equating this to the coefficient from the Maclaurin series formula:

$$\frac{f^{(5)}(0)}{5!} = \frac{1}{2}$$

Solving for the derivative:

$$f^{(5)}(0) = 5! \cdot \frac{1}{2} = 120 \cdot \frac{1}{2} = 60$$

The exact value is:

$$60$$

Physics & Engineering

4.1 Mechanics

Projectile Motion with Viscous Drag

A projectile of mass $m = 200$ g is launched in a viscous medium at an angle $\theta = 60^\circ$ with the horizontal, with an initial velocity $v_0 = 270$ m s^{-1}. It experiences a viscous drag force $\vec{F} = -c\vec{v}$ where $c = 0.1$ kg s^{-1}. The projectile hits a vertical wall after $t = 2$ s. Taking $e = 2.7$, find the horizontal distance of the wall from the point of projection.

Solution: We begin by isolating the horizontal motion of the projectile. The only force acting on the projectile in the horizontal direction is the horizontal component of the linear viscous drag force. According to Newton's Second Law:

$$m\ddot{x} = -c\dot{x} \implies \ddot{x} + \frac{c}{m}\dot{x} = 0$$

Let $k = \frac{c}{m}$. Given $m = 200$ g $= 0.2$ kg and $c = 0.1$ kg s^{-1}, we find $k = \frac{0.1}{0.2} = 0.5$ s^{-1}. This is a first-order linear differential equation in velocity $\dot{x}(t) = v_x(t)$:

$$\dot{v}_x = -kv_x$$

Integrating gives the velocity as a function of time:

$$v_x(t) = v_{x0}e^{-kt}$$

The initial horizontal velocity is $v_{x0} = v_0\cos(\theta) = 270\cos(60^\circ) = 270(0.5) = 135$ m s^{-1}. To find the horizontal displacement $x(t)$, we integrate the velocity function from time 0 to t:

$$x(t) = \int_0^t 135e^{-0.5\tau}\,d\tau = 135\left[\frac{e^{-0.5\tau}}{-0.5}\right]_0^t = -270\left(e^{-0.5t} - 1\right) = 270\left(1 - e^{-0.5t}\right)$$

We are given that the projectile hits a vertical wall at $t = 2$ s. We substitute this into our displacement equation to find the distance of the wall:

$$x(2) = 270\left(1 - e^{-0.5(2)}\right) = 270\left(1 - e^{-1}\right) = 270\left(1 - \frac{1}{e}\right)$$

The problem instructs us to use the approximation $e = 2.7$:

$$x(2) = 270\left(1 - \frac{1}{2.7}\right) = 270\left(1 - \frac{10}{27}\right) = 270\left(\frac{17}{27}\right)$$

$$x(2) = 10 \times 17 = 170 \text{ m}$$

The horizontal distance to the wall is:

$$170 \text{ m}$$

4.2 Quantum Mechanics

Infinite Square Well Expansion Probability

A particle is initially in the ground state of a one-dimensional infinite square well occupying $0 < x < L$. At $t = 0$, the right wall is moved instantaneously from $x = L$ to $x = 2L$. Determine the probability that the particle is found in the ground state of the enlarged well immediately after expansion.

Solution: The initial state of the particle $\psi(x)$ is the ground state of the original well of width L:

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \quad \text{for } 0 < x < L$$

and $\psi(x) = 0$ everywhere else. Because the expansion is instantaneous (sudden approximation), the wavefunction $\psi(x)$ does not change during the expansion process. The new ground state of the enlarged well (width $2L$) is:

$$\phi_1(x) = \sqrt{\frac{2}{2L}} \sin\left(\frac{\pi x}{2L}\right) = \sqrt{\frac{1}{L}} \sin\left(\frac{\pi x}{2L}\right) \quad \text{for } 0 < x < 2L$$

The probability P of finding the particle in this new ground state is the absolute square of the overlap integral (the inner product) between the initial state and the new state: $P = |\langle \phi_1 | \psi \rangle|^2$. We compute the coefficient $c_1 = \langle \phi_1 | \psi \rangle$:

$$c_1 = \int_0^{2L} \phi_1^*(x)\psi(x)\, dx = \int_0^L \sqrt{\frac{1}{L}} \sin\left(\frac{\pi x}{2L}\right) \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) dx$$

(Note the upper limit of the integral is L because $\psi(x) = 0$ for $x > L$).

$$c_1 = \frac{\sqrt{2}}{L} \int_0^L \sin\left(\frac{\pi x}{2L}\right) \sin\left(\frac{\pi x}{L}\right) dx$$

Using the product-to-sum trigonometric identity $\sin(A)\sin(B) = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$c_1 = \frac{\sqrt{2}}{2L} \int_0^L \left[\cos\left(\frac{\pi x}{L} - \frac{\pi x}{2L}\right) - \cos\left(\frac{\pi x}{L} + \frac{\pi x}{2L}\right)\right] dx$$

$$c_1 = \frac{\sqrt{2}}{2L} \int_0^L \left[\cos\left(\frac{\pi x}{2L}\right) - \cos\left(\frac{3\pi x}{2L}\right)\right] dx$$

Evaluating the integrals:

$$c_1 = \frac{\sqrt{2}}{2L} \left[\frac{2L}{\pi} \sin\left(\frac{\pi x}{2L}\right) - \frac{2L}{3\pi} \sin\left(\frac{3\pi x}{2L}\right)\right]_0^L$$

Substituting the upper limit $x = L$ (the lower limit yields 0):

$$c_1 = \frac{\sqrt{2}}{2L} \left[\frac{2L}{\pi} \sin\left(\frac{\pi}{2}\right) - \frac{2L}{3\pi} \sin\left(\frac{3\pi}{2}\right)\right]$$

Since $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$:

$$c_1 = \frac{\sqrt{2}}{2L}\left[\frac{2L}{\pi}(1) - \frac{2L}{3\pi}(-1)\right] = \frac{\sqrt{2}}{2L}\left(\frac{2L}{\pi} + \frac{2L}{3\pi}\right) = \frac{\sqrt{2}}{2L}\left(\frac{8L}{3\pi}\right) = \frac{4\sqrt{2}}{3\pi}$$

The transition probability is $P = |c_1|^2$:

$$P = \left(\frac{4\sqrt{2}}{3\pi}\right)^2 = \frac{16 \times 2}{9\pi^2} = \frac{32}{9\pi^2}$$

The exact probability is:

$$\frac{32}{9\pi^2}$$

4.3 Electromagnetism

Electron Acceleration in a Parallel-Plate Capacitor

An electron is released from rest at the negative plate of a parallel-plate capacitor. The potential difference between the plates is $U = 200$ V, and the distance between the plates is $d = 0.02$ m. The mass of the electron is $m = 9.11 \times 10^{-31}$ kg, and its charge magnitude is $e = 1.60 \times 10^{-19}$ C. Determine the magnitude of the acceleration a (in units of 10^{15} m s^{-2}) of the electron.

Solution: First, we find the magnitude of the uniform electric field E between the parallel plates of the capacitor. The electric field is the spatial gradient of the electric potential:

$$E = \frac{U}{d} = \frac{200\text{ V}}{0.02\text{ m}} = 10,000\text{ V m}^{-1}$$

The electron experiences an electric force F directed opposite to the field vector, but we are only interested in the magnitude. The force is given by the product of the elementary charge e and the electric field E:

$$F = eE = (1.60 \times 10^{-19}\text{ C}) \times (10^4\text{ V m}^{-1}) = 1.60 \times 10^{-15}\text{ N}$$

Using Newton's Second Law of Motion, we can find the uniform acceleration a of the electron by dividing the electric force by the electron's invariant mass m:

$$a = \frac{F}{m} = \frac{1.60 \times 10^{-15}\text{ N}}{9.11 \times 10^{-31}\text{ kg}}$$

$$a = \left(\frac{1.60}{9.11}\right) \times 10^{16}\text{ m s}^{-2}$$

Performing the arithmetic division $\frac{1.60}{9.11} \approx 0.17563$:

$$a \approx 0.17563 \times 10^{16}\text{ m s}^{-2} = 1.7563 \times 10^{15}\text{ m s}^{-2}$$

Rounding to an appropriate number of significant figures, consistent with the given inputs (which have 3 sig figs):

$$a \approx 1.76 \times 10^{15} \text{ m s}^{-2}$$

The magnitude of the acceleration a in units of 10^{15} m s^{-2} is:

$$1.76$$